

Lesson 5 · Mixed Practice

Learning intention

Choose the right calculus method from the wording of a question, and write Merit and Excellence answers under time pressure.

Success looks like

- I can name the method before I start writing.
- I can link two steps and justify a maximum in a sentence.
- I can build a formula from words and optimise it.

Slide 1 · Lesson 5 · Mixed Practice

Teaching line, not a question slide.

Timing: Do Now 10 min, three rounds about 20 min each, timed practice sheet 30 min, wrap-up 5 min.

The handout is the exam guide: point to it throughout.

Do Now

1 Differentiate $y = (3x - 1)(x + 2)$.

2 Find the turning point of $y = 2x^2 - 12x + 7$ and state its nature.

3 $f'(x) = 8x - 3$ and $f(1) = 6$. Find $f(x)$.

4 $f'(x)$ changes from negative to positive at $x = -2$. What happens on the graph of f at $x = -2$?

Slide 2 · Do Now

Q1 $\frac{dy}{dx} = 6x + 5$

Q2 $(3, -11)$, minimum

Q3 $f(x) = 4x^2 - 3x + 5$

Q4 a minimum (a turning point)

Short solution:

Q1 expand: $3x^2 + 5x - 2$

Q2 $4x - 12 = 0$; $y = 18 - 36 + 7$; $y'' = 4 > 0$

Q3 $4x^2 - 3x + c$; $4 - 3 + c = 6$

Watch for:

Q3 forgetting to find c

Diagnostic:

one question per lesson (1, 2, 4, 3): a wrong answer shows which lesson to revisit in round 1

Q4 wrong: Lesson 3 needs a quick recap with the translation table

The one idea

read the words → **name the method** → **write the calculus**

“Use calculus” means the derivative or anti-derivative must be in your working.

A calculator answer alone shows no method.

Slide 3 · The one idea

Teaching line, not a question slide.

The method chooser on handout page 1 is the reference for round 1.

Which method?

Name the method for each. Do not solve.

1. Find the gradient of the curve at $x = 4$.
2. Find the equation of the tangent at $x = -1$.
3. Find the maximum height of the ball.
4. The curve passes through $(2, 5)$; find $f(x)$ given $f'(x)$.
5. For what values of x is f decreasing?
6. Given the acceleration, find how far the car travels.

Slide 4 · Which method?

1 differentiate, substitute $x = 4$ into $f'(x)$

2 point from f , gradient from f' , then $y - y_1 = m(x - x_1)$

3 $h' = 0$, solve for t , substitute into h , justify with h''

4 anti-differentiate, $+c$, substitute $(2, 5)$

5 solve $f'(x) < 0$: an interval

6 anti-differentiate twice ($a \rightarrow v \rightarrow s$), a constant each time

Short solution:

say the method aloud; the student points to the matching row of the chooser

Watch for:

item 6 is distance, so check whether the car turns round

Practice

- 1 Find the gradient of $y = 2x^3 - 5x + 4$ at the point where $x = -2$.
- 2 $f'(x) = 12x^2 - 6$ and $f(1) = 3$. Find $f(x)$.
- 3 A particle has velocity $v = 6t - t^2 \text{ m s}^{-1}$. Find its acceleration at $t = 2$.

Slide 5 · Practice

Q1 19

Q2 $f(x) = 4x^3 - 6x + 5$

Q3 2 m s^{-2}

Short solution:

Q1 $\frac{dy}{dx} = 6x^2 - 5; 6(4) - 5$

Q2 $4x^3 - 6x + c; 4 - 6 + c = 3$

Q3 $a = 6 - 2t$

Watch for:

Q1 $6(-2)^2 = 24$, not -24

Q3 differentiate v once, not twice

Merit: link two steps

Two unknowns $\rightarrow$ two equations. A value given $\rightarrow$ solve for the time first.

4 The curve $y = ax^3 + bx$ has a turning point at $(1, -4)$. Find a and b .

equation 1: the point is on the curve

equation 2: the gradient there is 0

Slide 6 · Merit: link two steps

Q4 $a = 2, b = -6$

Short solution:

Q4 point: $a + b = -4$; turning point: $3a + b = 0$; subtract: $2a = 4$

Watch for:

using only “gradient = 0” and guessing the second letter

Merit marker: both conditions written as labelled equations

Practice

- 1 The curve $y = x^2 + px + q$ has a turning point at $(2, 5)$. Find p and q .
- 2 The number of visitors at a festival is $V = 600t - 25t^2$ after t hours ($0 \leq t \leq 24$). Find the rate of change of visitors when $V = 3200$.
- 3 Find the maximum value of $f(x) = 3 + 12x - 2x^3$ for $x \geq 0$, and justify that it is a maximum.

Slide 7 · Practice

Q1 $p = -4, q = 9$

Q2 200 and -200 visitors per hour

Q3 $3 + 8\sqrt{2} \approx 14.3$ at $x = \sqrt{2}$

Short solution:

Q1 gradient: $4 + p = 0$; point: $4 + 2p + q = 5$

Q2 $25t^2 - 600t + 3200 = 0$, so $t = 8$ or 16 ; $V' = 600 - 50t$

Q3 $f' = 12 - 6x^2 = 0, x = \sqrt{2}; f'' = -12x < 0$

Watch for:

Q2 two times give two rates: increasing at $t = 8$, decreasing at $t = 16$

Q3 reject $x = -\sqrt{2}$ because $x \geq 0$

Excellence: build the model

Write it in this order

1. Name the variables.
2. Constraint $\rightarrow$ one variable.
3. Differentiate, $= 0$, solve; reject impossible values.
4. Justify; answer in a sentence with units.

5 An open box has a square base of side x cm and height h cm. Its outside surface area is 108 cm^2 . Find the maximum volume.

Slide 8 · Excellence: build the model

Q5 108 cm^3 ($x = 6, h = 3$)

Short solution:

$$\text{Q5 } x^2 + 4xh = 108, \text{ so } h = \frac{108 - x^2}{4x}$$

$$\text{Q5 } V = x^2h = 27x - \frac{1}{4}x^3; V' = 27 - \frac{3}{4}x^2 = 0, x = 6$$

$$\text{Q5 } V'' = -\frac{3}{2}x = -9 < 0$$

Watch for:

an open box has one base, not two: $x^2 + 4xh$

the answer is the volume, not x (L3)

Practice

1 A 48 cm piece of wire is cut into two parts. Each part is bent into a square. Where should the wire be cut so that the total area of the two squares is a minimum? Justify that it is a minimum.

2 For $y = x^3 - 3a^2x$, where $a > 0$ is a constant, show that the turning points are at $x = \pm a$, and find the difference between the maximum and minimum y -values in terms of a .

Slide 9 · Practice

Q1 cut it in the middle (24 cm each); minimum total area 72 cm^2

Q2 turning points at $x = \pm a$; difference $4a^3$

Short solution:

Q1 pieces x and $48 - x$: $A = \frac{x^2}{16} + \frac{(48 - x)^2}{16}$; $A' = \frac{x}{8} - \frac{48 - x}{8} = 0$; $A'' = \frac{1}{4} > 0$

Q2 $y' = 3x^2 - 3a^2 = 0$; $y(-a) = 2a^3$ (max), $y(a) = -2a^3$ (min)

Watch for:

Q1 the side of each square is a quarter of its piece

Q2 Excellence: keep a as a letter all the way; justify which is the maximum

Ways to lose marks

L1 Write the derivative before substituting.

L2 Rates from the derivative; values from the original.

L3 A point needs y ; answer the quantity asked.

L4 Justify every maximum and minimum.

L6 Meaning and units for every rate.

Slide 10 · Ways to lose marks

Teaching line, not a question slide.

Full list L1 to L6 in the handout; ask which one the student is most likely to make under time pressure.

Summary

1. Name the method from the wording first.
2. Merit: two conditions, two equations; value given, solve for t .
3. Excellence: variables, constraint, one variable, optimise, justify.
4. Every answer: working, sentence, units.

Slide 11 · Summary

Teaching line, not a question slide.

Ask the student to say each line back before looking at the screen.

Practice sheet · 30 minutes, timed

Exam conditions: no notes, calculator allowed.

- Question 1: four Achieved parts, about 12 minutes.
- Question 2: two Merit parts, about 10 minutes.
- Question 3: Excellence, about 8 minutes.

Attempt every part; write the derivative first.

Slide 12 · Practice sheet · 30 minutes, timed

Q1 (a) 4; (b) $y = 7x - 17$; (c) $c = 5$; (d) $0 < x < 4$

Q2 (a) $t = 0.5$ s and $t = 2$ s; (b) 36 m

Q3 $k = 192$

Short solution:

Q1 (a) $f'(x) = 4x^3 - 6x + 2$ at $x = -1$

Q1 (b) point (3, 4), $m = 4(3) - 5 = 7$

Q1 (d) $3x^2 - 12x = 3x(x - 4) < 0$

Q2 (a) $v = 12t^2 - 30t + 12 = 6(2t - 1)(t - 2)$

Q2 (b) $v \geq 0$ for $0 \leq t \leq 6$, so highest at $t = 6$: $s = 3t^2 - \frac{1}{3}t^3$

Q3 growing fastest when P' is a maximum: $P'' = 2k - 24t^2 = 0$ at $t = 4$

Watch for:

Q2 (b) the maximum is at the end of the interval, where $v = 0$

Q3 check $P''' < 0$ and $P'(4) > 0$

Full working: practice answer version (tutor's working, not an official mark scheme)

Homework

Homework sheet: 9 questions, about 60 minutes

- Questions 1 to 6: mixed Achieved and Merit revision.
- Questions 7 and 8: Excellence exam questions.
- Question 9: Extension, a general proof.

Optional extra: redo any earlier practice sheet question you got wrong.

Slide 13 · Homework

Teaching line, not a question slide.

Answers: homework answer version (tutor's working, not an official mark scheme).

Mark Questions 7 to 9 together with the student next lesson; they are the longest exam-style questions in the unit.